Differential Gene Expression and Pathway Analysis of RSV Infection Using the GSE80179 Dataset

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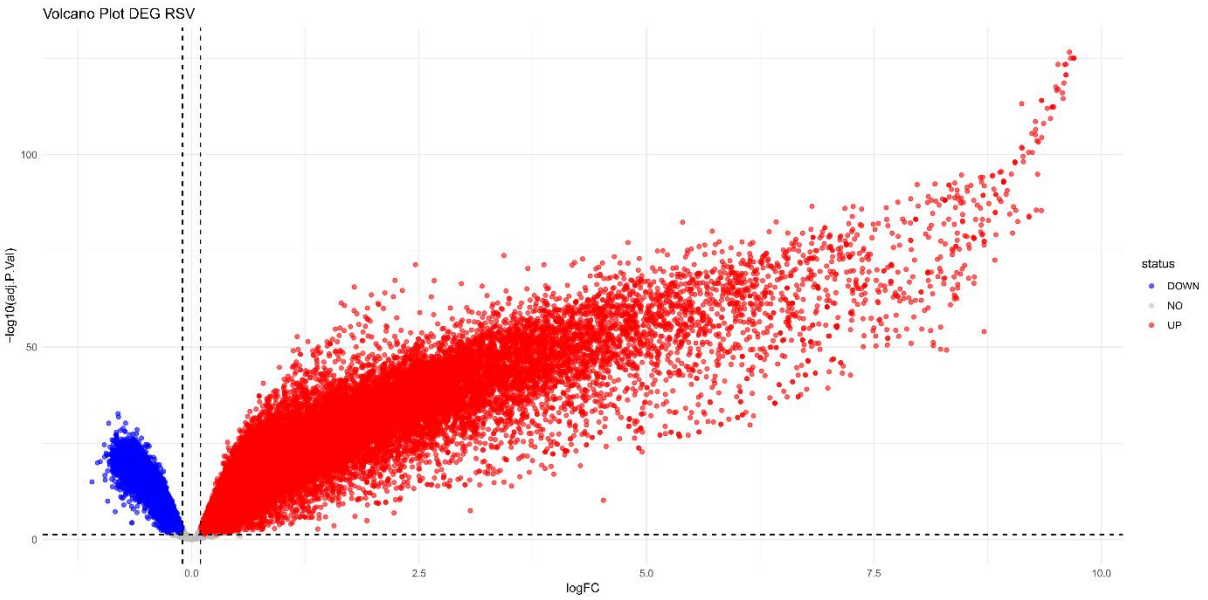
Respiratory viral infections can trigger significant changes in host gene expression, particularly in **pathways** related to **immune response, inflammation, and cellular metabolism.** Understanding these transcriptional changes is important for identifying molecular mechanisms involved in host-virus interactions. Publicly available transcriptomic datasets provide valuable resources for studying these responses. The **dataset GSE80179**, available in the Gene Expression Omnibus, contains gene expression profiles comparing respiratory syncytial virus (RSV)-infected samples with control samples. This **allows researchers to investigate differential gene expression** associated with **RSV infection** and explore the biological pathways involved in antiviral responses.

In this study, differential gene expression analysis was performed to identify genes that are significantly altered between RSV and control conditions. Functional enrichment analysis, including Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis, was conducted to understand the biological processes and molecular pathways associated with these differentially expressed genes.

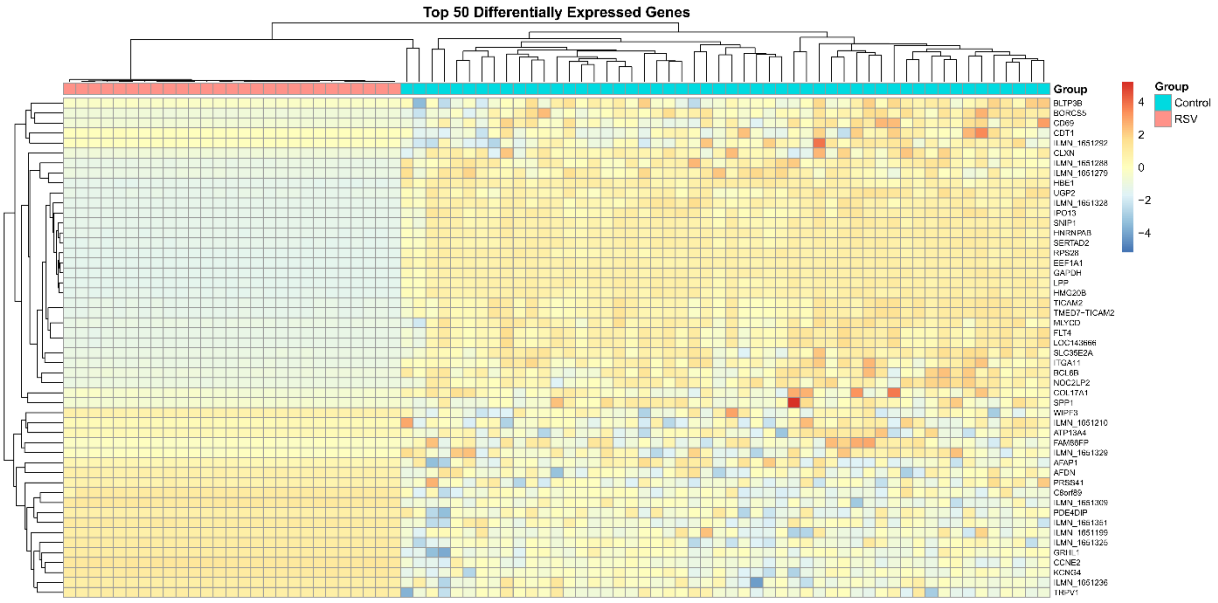
1. Data: Gene expression data were obtained from the **GEO** specifically the dataset **GSE80179**. The dataset was downloaded and processed using the **R** environment. The package **GEOquery** was used to retrieve the expression matrix and associated sample metadata.
2. DEG analysis: Gene expression analysis was performed using the **limma** package from the Bioconductor project. The expression matrix was extracted and samples were grouped according to their experimental condition (RSV infection vs. control).
3. Functional enrichment analysis: Functional enrichment analysis was performed to identify biological processes and pathways associated with the differentially expressed genes. Gene Ontology (GO) enrichment analysis was conducted using the **clusterProfiler** package. Pathway enrichment analysis was also performed using the **KEGG** database. Gene symbols were converted to Entrez Gene IDs using the **org.Hs.eg.db** to ensure compatibility with KEGG pathway analysis.
4. Data visualization: Several visualization methods were used to explore the transcriptomic data. Distribution of gene expression values across samples was evaluated using boxplots and density plots. Dimensionality reduction was performed using **UMAP** to visualize the clustering of samples based on global gene expression patterns. Additionally, volcano plots and heatmaps were generated to highlight significantly differentially expressed genes and their expression patterns across samples.

After preprocessing and statistical analysis using the **limma** package, a large number of genes were identified as significantly differentially expressed between the two groups.

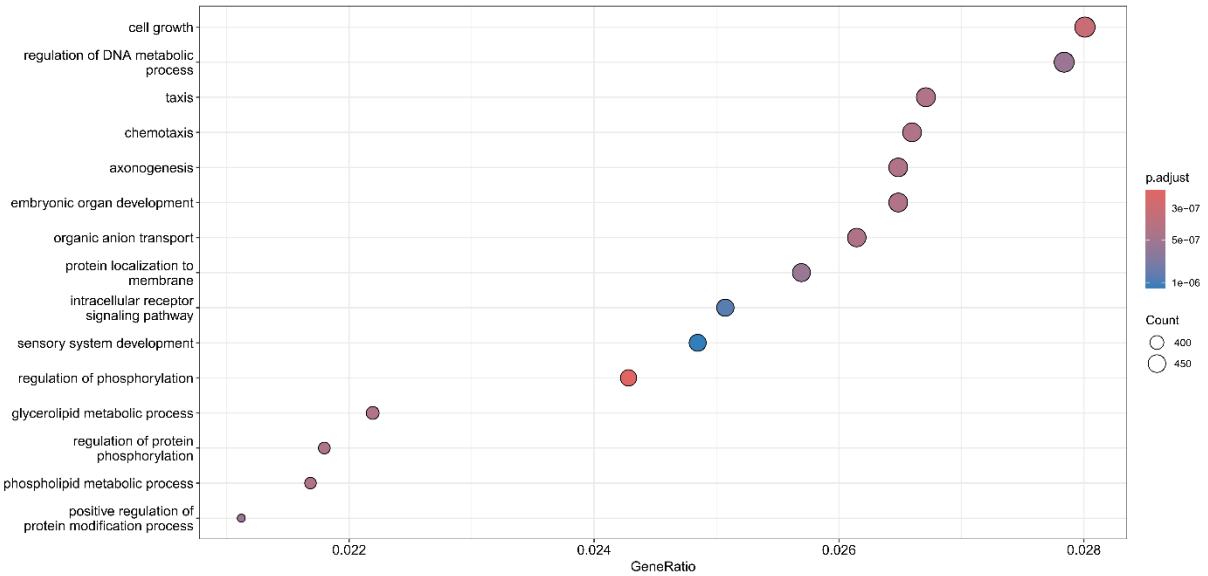
1. The **volcano plot illustrates** the distribution of **DEGs**. A substantial number of genes showed **significant upregulation** in the **RSV group**, indicated by the **large cluster of red** points with **positive log fold** change values. In contrast, **fewer genes were downregulated**, as shown by the **smaller number of blue points** on the **negative side** of the logFC axis. This pattern suggests that **RSV infection induces widespread transcriptional activation in host cells**.



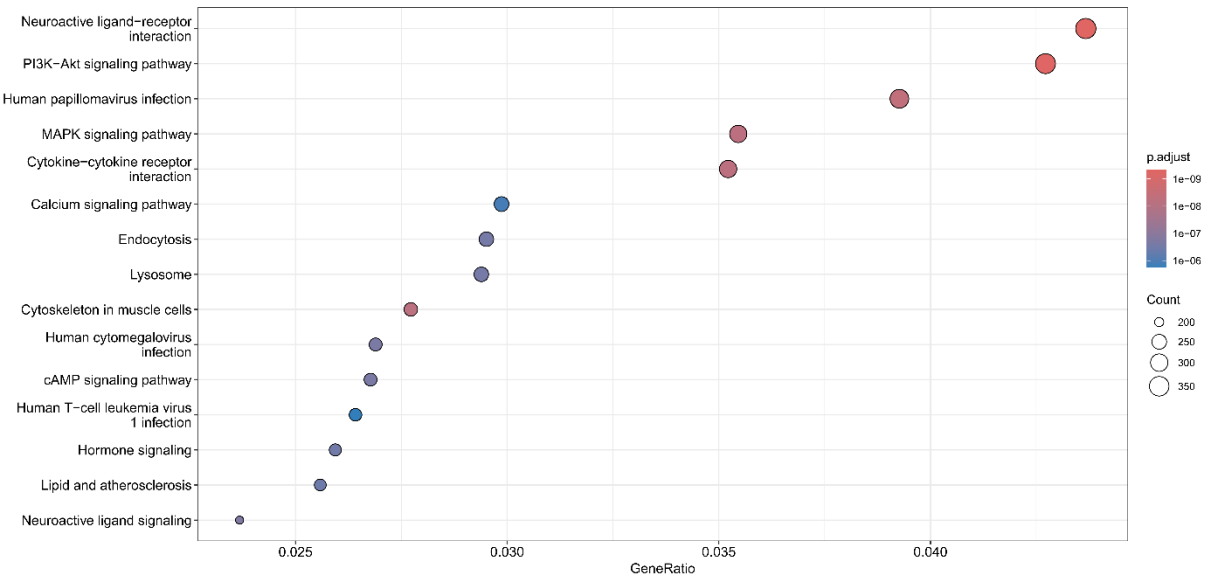
2. The **heatmap** of the **top 50** DEGs further demonstrates clear expression differences between the RSV and control groups. **Samples cluster** primarily **according** to the **experimental condition**, indicating that the identified **DEGs effectively distinguish infected samples from controls**. Many genes in the **RSV group show higher expression levels** compared with the control group, **reflecting virus-induced transcriptional** responses.



3. Gene Ontology enrichment analysis revealed several significantly enriched biological processes. Among the most prominent processes were **cell growth**, **regulation of DNA metabolic process**, **chemotaxis**, **taxis**, and **regulation of phosphorylation**. The **enrichment** of **chemotaxis** terms suggests that **immune cell recruitment and inflammatory** signaling **pathways** are activated during **RSV infection** (Kombe *et al.*, 2024). Additionally, processes related to **phosphorylation** regulation **indicate** the involvement of **signaling cascades** that control **cellular responses to viral infection**.



4. Pathway analysis using **KEGG** identified several significantly enriched signaling pathways. The most notable pathways include the **PI3K-Akt signaling pathway**, **MAPK signaling pathway**, and **cytokine–cytokine receptor interaction**. These pathways play essential roles in immune regulation, inflammation, and cell survival during viral infection (Muraro *et al.*, 2018; Huang *et al.*, 2022). Additional enriched pathways include **calcium signaling**, **endocytosis**, and **lysosome-related processes**, which are closely associated with **viral entry**, intracellular trafficking, and **immune defense mechanisms**. This reflects the **activation** of host **antiviral defense pathways** that are commonly shared among different viral infections.



In this study, transcriptomic data from the **GSE80179** were analyzed to identify genes and biological pathways associated with Respiratory Syncytial Virus infection. Differential expression analysis revealed thousands of genes that were significantly altered between RSV-infected and control samples, indicating a strong host transcriptional response to viral infection. Functional enrichment analyses demonstrated that many of these genes are involved in immune-related processes such as chemotaxis, cytokine signaling, and regulation of phosphorylation. Furthermore, pathway analysis highlighted several key signaling pathways, including PI3K-Akt and MAPK signaling pathways, which are known to play crucial roles in antiviral immune responses.

References

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